

Prolonged Pregnancy and Induction of Labour

Definitions

- Post-term pregnancy (prolonged pregnancy):
- Is defined as a pregnancy progressing to 42 weeks (294 days) of gestation or more from the first day of the last menstrual period (WHO&FIGO definition).
- The terms prolonged and post-term pregnancy are often interchanged but refers to the same condition.
- Although 42weeks is the current threshold for post-term pregnancy, many clinicians currently use the term prolonged pregnancy to describe pregnancies of 41week' gestation and beyond.
- Post-date: Is a pregnancy lasting beyond the estimated date of delivery (EDD) at 40 weeks.

Post-maturity syndrome

- This term describe the pathologic syndrome seen in post-term fetuses in which the fetus experiences placental insufficiency and consequent IUGR.
- It affect 20 % of cases of post-term pregnancy and is associated with:
 1. Meconium -stained amniotic fluid.
 2. Oligohydramnios.
 3. Fetal distress.
 4. Evidence of loss of subcutaneous fat.
 5. Dry, cracked skin.

All these features related to placental insufficiency.

Incidence

- The incidence of post-term pregnancy is about 10% depending on menstrual dating (1st day of the last menstrual period).
- While use of U/S to establish gestational age lowers the incidence of post-term pregnancies from 10 % to 2-5 %.
- The available evidences are strongly in support that dating by early ultrasonography alone is the most accurate method for calculating gestational age & predicting EDD.
- So now it is recommended that all pregnant ladies (and certainly those who do not have regular menses), should have an ultrasound examination for gestational age determination, prior to 16 weeks gestation.

Etiologic Factors

- The most frequent cause is an error in dating.
- When truly exists, the cause is usually unknown.
- But there many risk factors like:
 - Anencephaly.
 - Placental sulphate deficiency.
 - Fetal adrenal hypoplasia.
 - Male fetus.
 - Previous post-term pregnancy.
 - Nulliparity.
 - White race.
 - Genetic predisposition.

Risks of prolonged pregnancy

Risks to fetus:

1. Increased risk of intra-partum fetal death (4 times) & neonatal death (3 times) in infants born after 42 weeks compared with term fetus.
2. Oligohydramnios, meconium release and IUGR (post-maturity syndrome).
3. Increased risk of meconium aspiration by the fetus.
4. Macrosomia (birth weight > 4kg)
 - Macrosomia is more common in post-term than term pregnancies .
 - macrosomia by itself is not bad but it affect the baby adversely by causing:
 - difficult labour, Shoulder dystocia(difficult delivery of shoulders) with resultant risks of orthopedic or neurologic injury.

Maternal risks

1. Labour dystocia.
2. Severe perineal injury related to macrosomia.
3. Doubling in the rate of caesarean delivery.
4. A source of extreme anxiety for the pregnant woman.

Are there any interventions to prevent prolonged pregnancy?

- Sweeping the membranes performed as a general policy from 39 weeks onwards have been recommended to encourage spontaneous labour at term and decreased the frequency of prolonged pregnancy and the need for induction of labour.
- This procedure involve passing a finger through the cervical os, sweeping it around the internal surface of the cervix and gently pushing the membrane surface away from the cervix.
- It is associated with increase circulating prostaglandins.
- Increased likelihood of spontaneous labour within 48 hours and reduce the need for IOL.

Management of prolonged pregnancy

-Although prolonged pregnancy considered at 42weeks, most women seeks advice once she passed her EDD.

Options are either delivery by induction of labour or expectant management depend on:

1. Gestational age.
2. Absence/presence of maternal risk factors and / or
3. Evidence of fetal compromise.
4. Maternal preferences .

- Successful management depends on effective counselling of women and their full involvement in the decision making process.

Antenatal fetal monitoring of prolonged pregnancy

A modified biophysical profile test consisting of a:

- Non stress test (NST) by Cardiotocography +
- U/S assessment of amniotic fluid index.
- have been shown to be sensitive test to monitor the fetus in prolonged pregnancy.

Admit the woman for immediate delivery if there is:

1. Maternal risk factors e.g.: pre-eclampsia, DM, Rh – incompatibility...ect.
2. Evidence of fetal compromise:
 - Decrease fetal movement.
 - Abnormal NST (by CTG).
3. Oligohydramnios by U/S (AFI<5cm).

For Healthy, uncomplicated pregnancy, normal fetal movement and amniotic fluid: offer follow up till 41week.

Management at 41 weeks gestation

Offer the woman induction of labour (IOL) even in healthy uncomplicated pregnancy .

- ▶ After counseling the woman of risks/ benefits of labour induction versus expectant management,
 - ▶ Because several large multicenter randomized studies reported -Favourable outcomes with routine induction of labour (IOL) at 41 weeks of gestation.
- The risk of fetal distress from utero-placental insufficiency due to prolonged pregnancy can be reduced by IOL at 41weeks.

If mother declines IOL at 41weeks ,

Then provide expectant management with:

- Daily fetal movement counts
- Twice weekly non stress test (NST) and amniotic fluid index (AFI) up to 42 weeks.

Any time the NST or AFI is become abnormal:

- Deliver the woman immediately.

**Once the woman reach 42
weeks we have to deliver her
even if NST and AFI are still
normal.**

Management during labour and delivery

- Consider amniotomy to diagnose meconium.
- If meconium is present then consider risk of meconium aspiration ,so continuous electronic fetal monitoring (by CTG) is recommended.
- Be prepared for shoulder dystocia and neonatal resuscitation at delivery.

Induction of labour (IOL):

- The process of artificially initiating uterine contractions, prior to their spontaneous onset, with the intention to effect progressive effacement and dilatation of the cervix and, ultimately delivery of the baby.
- Induction of labour generally refers to procedures performed in the 3rd trimester.
- Before 24 weeks it is called termination of pregnancy.

Indications for Induction of labour:

- The purpose of induction is to achieve benefits to the health of the mother and / or fetus which must exceed that to be gained by continuing the pregnancy.

Indications for Induction of labour:

- Prolonged pregnancy (usually done after 41 completed weeks).
- Pre-labour spontaneous rupture of membranes (PROM).
- Maternal diseases: Diabetes mellitus, renal disease, autoimmune diseases, hypertension..ect.
- Pregnancy related conditions:
 - Pre-eclampsia.
 - Intrahepatic cholestasis of pregnancy.
 - Recurrent APH.
- Fetal causes:
 - IUGR
 - intrauterine fetal death(IUFD).
 - Oligohydramnios.
 - Rh- isoimmunization.
 - Macrosomia (big fetus >4KG).
 - Fetal anomalies incompatible with life.
- Social reason.

Contraindications to Induction of labour:

- Major degree placenta praevia.
- Severe IUGR & fetal compromise.
- Sever cephalo-pelvic disproportion.
- Deteriorating maternal condition with major antepartum haemorrhage, pre-eclampsia or cardiac disease may favour caesarean delivery.
- Transverse lie .
- Breech presentation is a relative contraindication to IOL.
- Previous 2 and more C/S and myomectomy,
- Previous one C/S (relative contraindication) the woman should be informed about risk of rupture of uterus.

The 'ripeness' of the uterine cervix:

- The process of softening, shortening and partial dilation of the cervix usually takes place in the days or weeks prior to the onset of labour, but the timing of this process is variable.
- An 'unripe' cervix is one which has undergone minimal change and is more resistant to attempts at IOL.
- Unripe' cervix is termed unfavourable for IOL
- If labour is induced before this process has occurred, the induction process will take longer time or fail.

The Bishop score:

-Bishop in 1964 produced a clinical scoring system of the cervix to quantify the degree of ripeness of cervix prior to IOL

	0	1	2	3
Dilatation of cervix (cm)	0	1-2	3-4	5 or more
Consistency of cervix	Firm	Medium	Soft	
Length of cervical canal (cm)	>2	2-1	1-0.5	<0.5
Position of cervix	Posterior	Central	Anterior	
Station (cm above ischial spines)	-3	2-	-1or 0	Below the spine

Bishop score:

-High scores ≥ 6 indicate a 'favourable' cervix : associated with an easier, shorter induction process that results mostly in successful vaginal delivery.

-Low scores <6 indicate an 'unfavourable' cervix :

-IOL will take long time and mostly fail.

-Such women needs IOL by drugs that induce cervical ripeness and uterine contractions at same time (like PGs).

Complications of IOL:

- Increase pain and need for analgesia.
- Uterine hyper-stimulation.
- Failure leading to Caesarean section.
- Fetal distress.
- intrauterine infection.
- Uterine rupture (very rarely).
- Post-partum haemorrhage.
- Precipitate/dysfunctional labour.
- Increased risk of operative vaginal delivery and birth trauma.
- Anxiety if the induction is protracted or unsuccessful.

Pre-requisites:

- Establish indication clearly.
- Informed consent.
- Conformation of gestational age.
- Assessment of fetal size, lie & presentation.
- Do pelvic assessment & assessment of Bishops score.

Methods of induction of labour:

1. Natural:

- Breast/nipple stimulation.
- Sexual intercourse.
- Acupuncture/acupressure.
- Membrane sweeping

2. Mechanical:

- Balloon catheters
- Laminaria tents.
- Synthetic osmotic dilators.
- Artificial rupture of membranes (amniotomy)

3. Chemical:

A. Non- hormonal

- Herbs , evening primrose oil
- Enemas
- Castor oil (risk of meconium)

B. Hormonal

- Prostaglandin E2
- Misoprostol.

- Oxytocin.
- Mifepristone.
- Relaxin.
- Nitric oxide donor.

Amniotomy (artificial rupture of the membranes) (ARM)

-Procedure :

- The FHR is recorded before the procedure.
- A pelvic examination is performed to evaluate the cervix & station of the presenting part.
- The presenting part should be well fitted to the cervix.
- The membranes are identified and a Kocher is inserted through the cervical os by sliding it along the hand & fingers & membranes are ruptured.
- The nature of the amniotic fluid is recorded (clear, bloody, thick or thin, meconium).
- The FHR is recorded after the procedure.

Risks of amniotomy (ARM):

- 1- Prolapsed of the umbilical cord (0.5%)**
- 2- Chorioamnionitis: Risk increases with prolonged induction-delivery interval**
- 3- Postpartum hemorrhage: Risk is doubled compared with women with spontaneous onset of labor**
- 4- Rupture of vasa previa .**
- 5- fetal injury.**

Pharmacological methods for IOL

Prostaglandins (PGs): PGE2 (dinoprostone)

- These are long chain fatty acids derived from arachidonic acid via the cyclo-oxygenase pathway.
- They exert a powerful effects on the cervix and the myometrium at all stages of pregnancy.
- They modify the ground substance within the cervix increasing its compliance (induce ripeness), they also stimulate uterine contraction and thus inducing labour.
- They are particularly useful for labour induction where the cervix is unfavourable.

The most common route of administration is vaginal, and tablets, suppositories, gels and pessaries have been developed.

-It can be given as intracervical but it is more invasive.

-A limiting factor for the use of prostaglandin E2 preparations in many countries has been the cost.

Misoprostol:

- **Misoprostol (Cytotec)** is a synthetic PGE1 analog that has been found to be a safe and inexpensive agent for cervical ripening and labour induction.
- Route of administration: Oral, vaginal and sublingual route for induction. Rectal route is used to prevent and treat postpartum hemorrhage.
- Bioavailability: Extensively absorbed from the GIT.
- Metabolism: De-esterified to prostaglandin F analogs.
- Half life: 20–40 minutes
- Excretion: Mainly renal 80%, remainder is fecal: 15%.
- Clinical trials indicate that the safe optimal dose and dosing interval for IOL is 25 mcg intra-vaginally every 4-6 hours.
- A maximum of 6 doses was suggested.

Misoprostol:

- Higher doses or shorter dosing intervals are associated with a higher incidence of side effects, especially hyper stimulation syndrome.
- Misoprostol should not be used in women with previous CS because of increased rates of uterine rupture.
- The use of misoprostol resulted in an over all lower incidence of CS.
- In addition, there appears to be a higher incidence of vaginal delivery within 24 hours of application and a reduced need for oxytocin augmentation.
- It is particularly useful for labour induction with unfavourable cervix.

Side effects of misoprostol:

1.Uterine hyper-stimulation (Tachysystole > 5contractions per 10min.)

2.Meconium-stained liquor.

- Certain myometrial stimulants may cross the placenta to stimulate fetal bowel smooth muscle and cause meconium passage.

3.Precipitate delivery:

-Precipitate delivery (labour < 2 h) has been described as a complication of misoprostol.

4. Rupture of the unscarred uterus.

5. maternal fever, chills, and GIT symptoms like nausea, vomiting, and diarrhea

Oxytocin:

- It is given by IV infusion using an automated pump.
- Oxytocin has many advantages: it is potent and easy to titrate, has a short half-life (one to five minutes) and is well tolerated.
- Oxytocin is administered as a variable dose infusion, titrated against uterine contractions.
- A typical dosage schedule : would be 5 IU in 500 NS at infusion rate of 1 mU /min, doubling the rate of infusion every 20–30min until adequate uterine contractions are achieved or a rate of 32 mU/min is reached.
- Once labour is established the infusion rate may be progressively reduced, as the myometrial sensitivity increases, to a rate of about 7 mU/min (maintenance).

Side effects of oxytocin:

- 1-Uterine hyper stimulation and subsequent FHR abnormalities.**
- 2-Abruptio placentae and uterine rupture.**
- 3-Water intoxication may occur with high concentrations of oxytocin infused with large quantities of intravenous solutions (due to the antidiuretic effect of oxytocin).**
- 4-A rapid intravenous injection of oxytocin may cause hypotension.**
- 5. Neonatal hyperbilirubinemia.**

Mifepristone:

- Mifepristone is an anti-progesterone agent which counteracts the inhibitory effect of Progesterone on the uterus.
- Women treated with mifepristone in a dose of 600 mg are more likely to have a favorable cervix and deliver within 48 to 96 hrs when compared with placebo and also they were less likely to undergo C.S.

Which drug to use for IOL

- The choice of drug depend on the Bishop score.
- Vaginal Prostaglandin E2 is the preferred drug when the cervix is unfavourable, it induce cervical ripening and stimulate uterine contractions..
- However, prostaglandins may equally be used when the cervix is favourable.
- Combination of amniotomy and oxytocin infusion for labour induction with unfavourable cervix are ineffective(oxytocin has no effect on the cervix) so they are used for IOL in women with favourable cervix.
- Misoprostol is equally effective to PGE2 and much cheaper and can be used for the same indications.

Thank you